

Insurance Charges Analysis Report

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1. Introduction

Health insurance companies determine premiums based on a range of personal and lifestyle factors, but the exact weight each factor carries is not always obvious. This report analyzes the Kaggle "Medical Cost Personal Datasets" (1,338 customer records) to identify which attributes most strongly influence insurance charges and to what extent.

The analysis addresses three main questions:

- What patterns exist in the data? — explored through summary statistics and visualizations covering age, BMI, smoking status, number of children, and region.
- Which factors most affect charges? — measured through correlation analysis and group comparisons (e.g., smokers vs. non-smokers).
- Can charges be predicted accurately? — tested by building regression models that estimate charges from customer attributes, evaluated using R^2 and Mean Absolute Error.

The findings demonstrate a complete, practical data analysis workflow — from raw data to working predictive models.

2. Dataset Overview

The dataset is the publicly available Kaggle "Medical Cost Personal Datasets" (insurance.csv). It contains 1,338 records and 7 columns:

Column	Type	Description
age	Numeric	Age of the primary beneficiary
sex	Categorical	Gender of the policyholder (male / female)
bmi	Numeric	Body Mass Index — a measure of body fat based on height and weight
children	Numeric	Number of dependents covered by the insurance plan
smoker	Categorical	Smoking status (yes / no)
region	Categorical	Residential area in the US (northeast, northwest, southeast, southwest)
charges	Numeric (Target)	Individual medical costs billed by health insurance, in USD

Key Characteristics

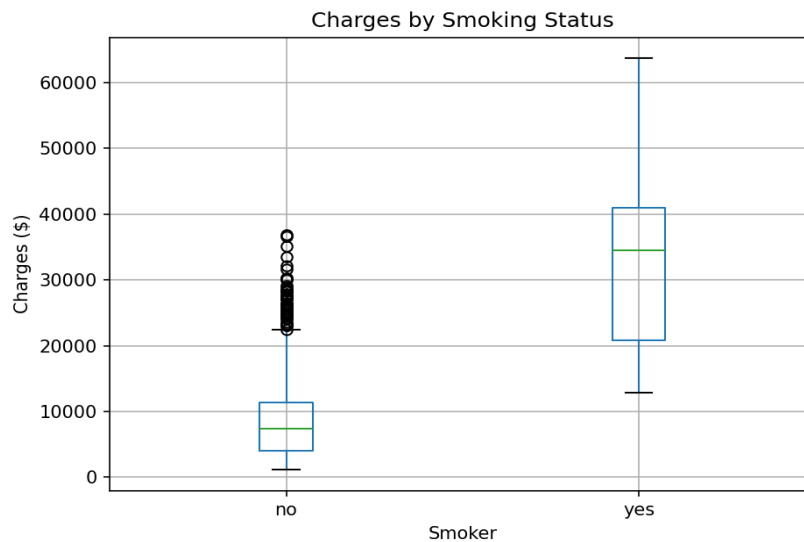
- Target variable: charges — this is what the models are trained to predict.
- Feature types: a mix of numeric (age, BMI, children) and categorical (sex, smoker, region) variables, requiring encoding before modeling.
- No missing values — the dataset is clean and complete, so no imputation was needed.
- Class imbalance in smoker: roughly 80% non-smokers vs. 20% smokers, worth noting since smoking status is later shown to be the strongest predictor of charges.

3. Exploratory Data Analysis

Before modeling, the dataset was explored to understand distributions, relationships, and any patterns that could explain variation in insurance charges.

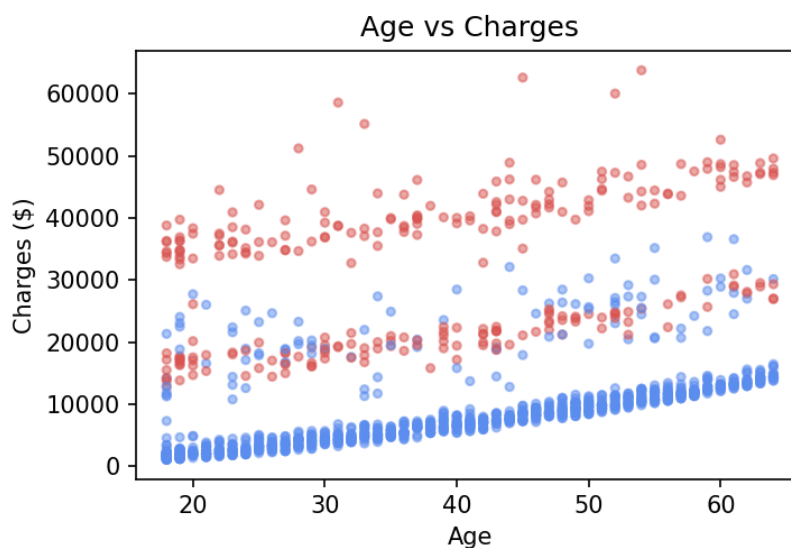
3.1 Charges by Smoking Status

Smoking status shows the clearest and largest difference in charges. Average charges are \$32,050.23 for smokers versus \$8,434.27 for non-smokers — roughly 3.8x more. The boxplot below shows smoking status separates customers into two clearly different charge ranges.



3.2 Age vs. Charges

There is a moderate positive correlation between age and charges ($r = 0.299$) — charges tend to rise as customers get older, consistent with increasing health risk over time. With smoker status colored in, smokers (red) sit as a higher band above non-smokers (blue), meaning age and smoking status affect charges largely independently.



3.3 BMI vs. Charges

BMI has a weaker positive correlation with charges ($r = 0.198$). Customers with BMI of 30 or more (obese range) average about \$15,552 in charges versus \$10,714 for those below 30, so BMI matters, but less than smoking status or age.

3.4 Region and Number of Children

Average charges by region were close to one another, from about \$12,347 (southwest) to \$14,735 (southeast), so region has only a minor effect. Number of children showed a similarly modest effect.

Summary of EDA Findings

Factor	Effect on Charges
Smoking status	Strong — largest single driver
Age	Moderate — steady positive trend
BMI	Weak-to-moderate positive
Region	Minimal
Children	Minimal

Key Statistics

Metric	Value
Rows in dataset	1,338
Avg. charges — smokers	\$32,050.23
Avg. charges — non-smokers	\$8,434.27
Correlation: age vs charges	0.299
Correlation: BMI vs charges	0.198

4. Methodology

Two models were trained to predict charges using age, sex, BMI, number of children, smoker status, and region: Linear Regression and Random Forest (200 trees). Categorical variables (sex, smoker, region) were one-hot encoded. The data was split 80/20 into training and test sets (random_state = 42).

5. Results

Performance on the held-out test set:

Model	R ² score	Mean Absolute Error
Linear Regression	0.784	\$4,181.19
Random Forest	0.864	\$2,563.57

Random Forest performed best, with an R² of 0.864 (explaining about 86% of the variation in charges) and an average prediction error of roughly \$2,564. Linear Regression reached an R² of 0.784, because the relationship between features and charges (especially smoking combined with BMI) is not purely linear.

6. Conclusion

- Smoking status is by far the strongest driver of insurance charges.
- Age and BMI both have a positive relationship with charges; region and children matter little.
- A non-linear model (Random Forest) predicts charges noticeably better than Linear Regression.

Future work could include hyperparameter tuning, gradient boosting (XGBoost), modeling the BMI × smoker interaction explicitly, and validating on additional real-world insurance data.